Bluestock Fintech: Mutual Fund Analytics Capstone

Final Project Report

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1. Executive Summary:

This report details the end-to-end development of a Mutual Fund Analytics Platform for Bluestock Fintech. Over the course of a 5-day sprint, raw financial data spanning 4.5 years was ingested, cleaned, and transformed into a robust relational database. Using Python and SQL, advanced exploratory data analysis (EDA) and quantitative risk/return metrics were computed to evaluate 40 specific mutual fund schemes. Finally, the data was materialized into an interactive 4-page Business Intelligence dashboard. The resulting platform empowers retail investors and internal analysts to make data-driven decisions based on historical performance, risk-adjusted returns, and demographic investment trends.

2. Data Overview:

The project utilized 10 distinct datasets encompassing over 87,000 rows of financial data.

- **Timeframe:** January 2022 to December 2025.
- **Scope:** 40 mutual fund schemes across various categories (Equity, Debt, Hybrid, Liquid).
- **Sources:** AMFI India (Public), mfapi.in, and NSE/BSE Public Data.
- **Key Domains:** Daily NAV history, investor transaction logs, portfolio holdings, monthly SIP inflows, and industry-wide AUM metrics.

3. ETL Pipeline & Architecture (Extract, Transform, Load):

A fully automated ETL pipeline was constructed using Python (Pandas) and SQLite.

- **Extraction & Cleaning:** Raw CSV files were ingested. Null values were handled via forward-filling (for time-series data) or zero-filling, and duplicate records were dropped. Data types were standardized (e.g., casting strings to datetime objects).
- **Data Modeling:** The cleaned data was loaded into a newly engineered SQLite database (bluestock_mf.db) utilizing a Star Schema architecture.
- **Database Schema:** * Dimension Table: dim_fund (Static fund metadata)

    - Fact Tables: fact_nav (Daily pricing), fact_transactions (Investor behavior), fact_performance (Aggregated metrics), and fact_portfolio_holdings (Underlying assets).

4. Exploratory Data Analysis (EDA) Findings:

Through Python-based visualizations (Matplotlib, Seaborn, Plotly), several key market insights were uncovered:

1. **Industry Consolidation:** SBI Mutual Fund demonstrates massive market dominance, peaking at ₹12.5 Lakh Crore AUM, significantly outpacing competitors.
2. **Retail Financialization:** Retail participation is accelerating rapidly, with monthly SIP inflows hitting an all-time high of ₹31,002 Crores in December 2025.
3. **Market Resilience:** Despite the mid-2024 market correction, NAV trends showed a swift, unified recovery, strongly correlated with the NIFTY 100 benchmark.
4. **Youth Participation:** The 26-35 age demographic forms the largest investor cohort, indicating a generational shift toward capital markets over traditional savings.
5. **Geographic Concentration:** SIP transaction volumes remain heavily concentrated in T30 (Top 30) cities, specifically within Maharashtra and Gujarat.

5. Fund Performance & Risk Analysis:

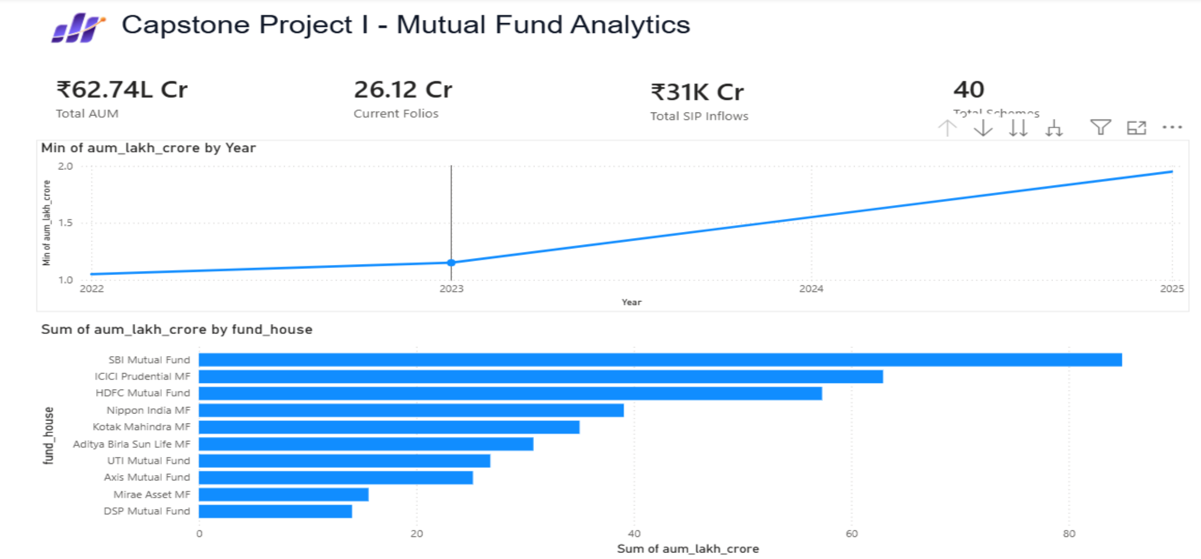
A quantitative analytics engine was built using numpy and scipy.stats to evaluate funds beyond simple returns.

- **Calculated Metrics:** 1Y/3Y/5Y CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown.
- **Composite Scorecard:** A weighted scoring model (0–100) was developed to rank funds:
    - 30% Weight: 3-Year CAGR
    - 25% Weight: Sharpe Ratio
    - 20% Weight: Alpha (Excess Return vs. Benchmark)
    - 15% Weight: Expense Ratio (Inverse)
    - 10% Weight: Maximum Drawdown (Inverse)

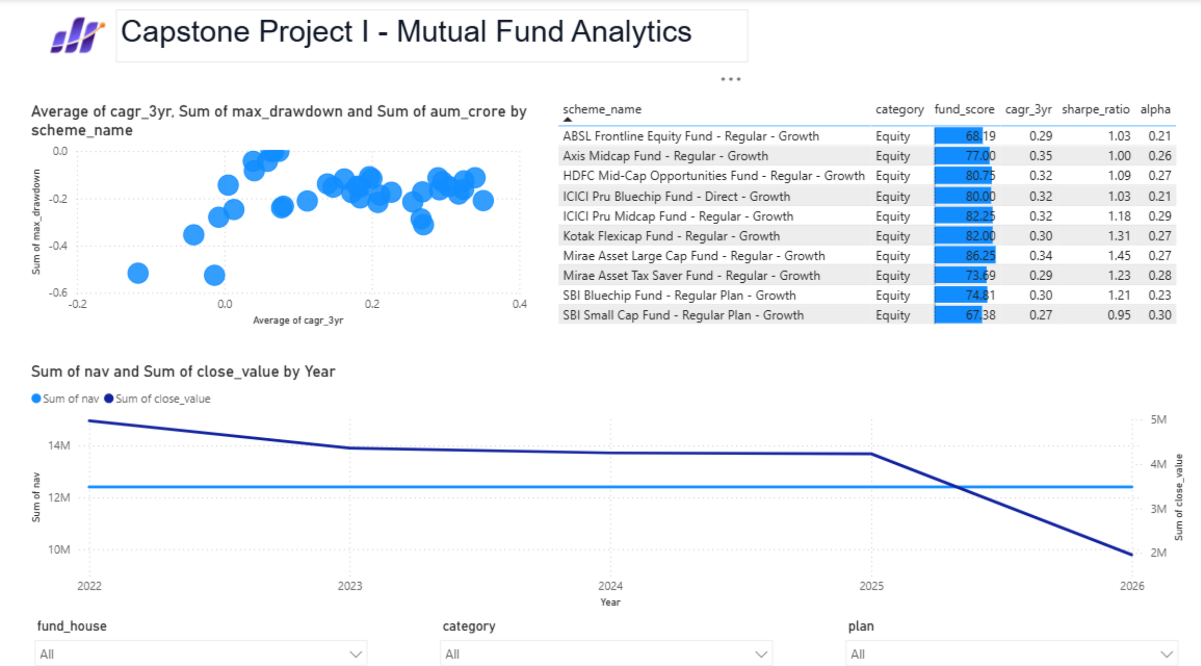
- **Outcome:** The scorecard successfully isolates funds that generate genuine Alpha rather than those simply riding broader market beta, allowing for highly optimized portfolio recommendations.

6. Dashboard Screenshots:

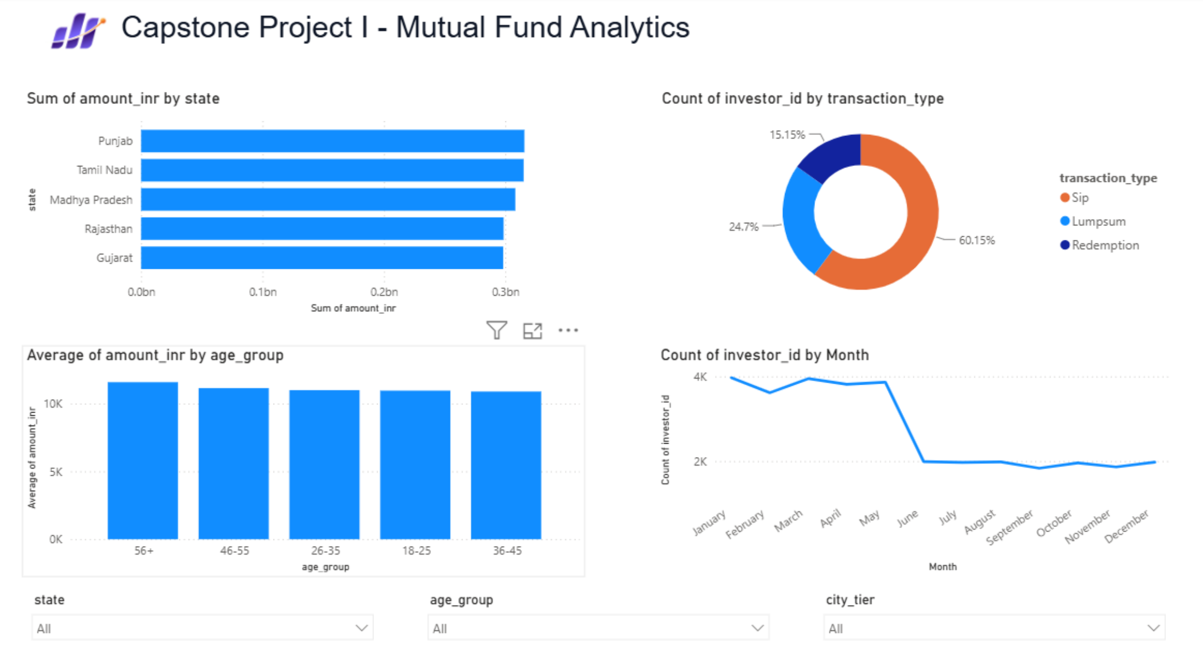
Page 1: Industry Overview:



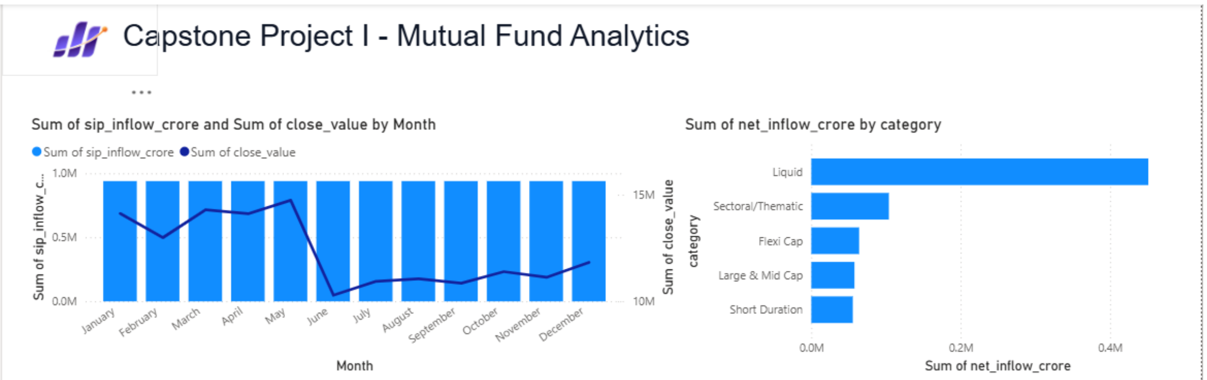
Page 2: Fund Performance:



Page 3: Investor Analytics:



Page 4: SIP & Market Trends:



category	January	February	March	April	May	June	July	August	September	October	November	December	Total
ELSS	516	437	500	466	553	472	471	499	537	537	571	521	**6080**
Flexi Cap	5603	6068	4767	4947	5529	4478	4869	5562	5397	6004	6111	4654	**63989**
Gilt	744	942	956	784	836	864	959	952	925	898	704	831	**10395**
Hybrid	2967	3360	2830	2955	3487	3163	3291	3684	3015	3314	3264	3538	**38868**
Large & Mid Cap	4816	5524	4243	4214	4368	4610	5023	5411	4528	4581	5556	4878	**57752**
Large Cap	2025	1925	2234	2413	2076	2519	2574	1940	1879	2255	1870	1923	**25633**
Liquid	33892	32374	38681	37537	41872	40486	34643	41952	35308	39091	40506	34933	**451275**
Mid Cap	4316	4819	5061	3897	5300	5047	4548	3899	4960	4106	4336	5023	**55312**
Sectoral/Thematic	7893	9215	8614	8052	8354	10030	9896	8360	8518	7680	7397	9820	**103829**
Short Duration	4752	4033	4886	4400	4833	4321	4170	4658	5327	4675	5316	4159	**55530**
Small Cap	4554	3534	4304	3533	4092	3535	3582	3376	4137	4444	3256	4249	**46596**
Value/Contra	1339	1400	1497	1328	1361	1386	1582	1308	1334	1595	1436	1414	**16980**
Total	**73417**	**73631**	**78573**	**74526**	**82661**	**80911**	**75608**	**81601**	**75865**	**79180**	**80323**	**75943**	**932239**

7. Strategic Recommendations:

Based on the data and performance analysis, the following actions are recommended for Bluestock Fintech:

1. **Target B30 Geographies:** Since current investments are highly concentrated in T30 states, there is massive untapped potential for marketing campaigns targeting Tier-2 and Tier-3 cities.
2. **Promote High-Alpha Funds:** Push funds that scored >80 on the custom scorecard to retail clients. These funds have mathematically proven their ability to generate risk-adjusted returns during the 2022-2025 volatile periods.
3. **Cater to the Youth:** The data shows young investors (18-35) drive SIP volume but have lower average investment amounts. Introduce "Micro-SIP" features (e.g., ₹500 minimums) to capture higher volumes of Gen-Z investors.

8. Limitations & Future Scope:

- **Data Scope:** The analysis is constrained to 40 specific mutual fund schemes. Expanding this to the full 1,900+ AMFI schemes would require transitioning from SQLite to a more robust cloud data warehouse (e.g., Snowflake, AWS Redshift).
- **Static Risk-Free Rate:** The Sharpe/Sortino ratios assume a static risk-free rate (6.5%). Future iterations should dynamically pull the 10-Year Government Bond yield.
- **Real-Time Integration:** The current ETL pipeline is batch-processed. Future development should implement live API connections (via mfapi.in) to provide real-time NAV updates to the dashboard.